INCH-POUND

MIL-PRF-39003/3G
22 July 2004
SUPERSEDING
MIL-PRF-39003/3F
17 May 1996

PERFORMANCE SPECIFICATION SHEET

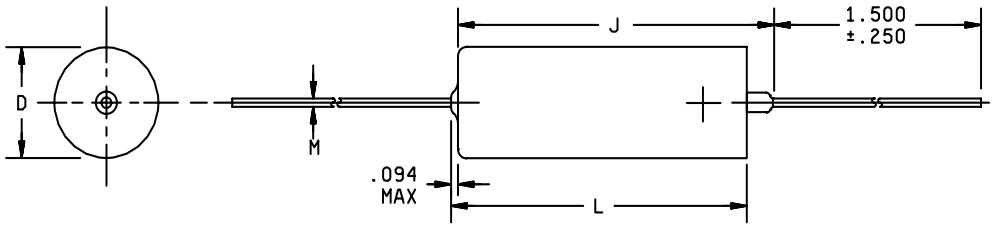
CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), ESTABLISHED RELIABILITY, STYLE CSR23

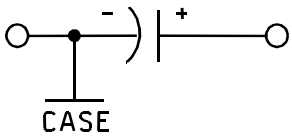
FAILURE RATE LEVELS "M", "P" AND "R" ARE INACTIVE FOR NEW DESIGN AFTER 2 FEBRUARY 1984.

FAILURE RATE LEVEL "S" IS INACTIVE FOR NEW DESIGN AFTER 15 AUGUST 1984.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39003.





CIRCUIT DIAGRAM

Inches	mm
.094	2.39
.250	6.35
1.500	38.10

FIGURE 1. Dimensions and configurations.

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Case size	Dimensions			
	L ± .031 (0.79)	D ± .016 (0.41)	M ± .002 (0.05)	J max
A	.286 (7.26)	.135 (3.43)	.020 (0.51)	.422 (10.72)
B	.474 (12.04)	.185 (4.70)	.020 (0.51)	.610 (15.49)
C	.686 (17.42)	.289 (7.34)	.025 (0.64)	.822 (20.88)
D	.786 (19.96)	.351 (8.92)	.025 (0.64)	.922 (23.42)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are given for general information only.
3. The case insulation shall extend .015 (0.38 mm) minimum beyond each end. However, when a shrink-fitted insulation is used, it shall lap over the ends of the capacitor body.
4. Lead length may be a minimum of 1-inch long for use in tape and reel automatic insertion equipment, when specified.

FIGURE 1. Dimensions and configuration - Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tubular, insulated.

Material: Metal (alloy unspecified).

Type of seal: Hermetic.

* Leads: Types N-32, N-52, W-32, or W-52 of MIL-STD-1276.

DC rated voltage: See table I. Above +85◦C voltage derating is required (see MIL-PRF-39003).

DC leakage (DCL): See table I.

Capacitance (Cap.) (nom): See table I.

Cap. tolerance: See table I.

Dissipation factor (DF): See table I.

Operating temperature: -55◦C to +125◦C.

Failure rate level: See table I.

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TABLE I. Style CSR23 characteristics and PIN's.

DC rated voltage	Capacitance (nom)	Capacitance tolerance	DC leakage at			Dissipation factor at -55°C through +125°C	Case size	PIN M39003/03- 1/ failure rate level for (%/1,000 hours)						
			+25°C	+85°C	+125°C			M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	μF	percent	μA	μA	μA	percent								
6	10	10	0.9	9	11	6	A	0101-	0201-	0301-	0401-	2001-	3001-	4001-
6	10	20	0.9	9	11	6	A	0102-	0202-	0302-	0402-	2002-	3002-	4002-
6	12	10	1	10	12.5	6	A	0103-	0203-	0303-	0403-	2003-	3003-	4003-
6	100	10	6	60	75	8	B	0104-	0204-	0304-	0404-	2004-	3004-	4004-
6	100	20	6	60	75	8	B	0105-	0205-	0305-	0405-	2005-	3005-	4005-
6	330	10	15	150	188	8	C	0106-	0206-	0306-	0406-	2006-	3006-	4006-
6	330	20	15	150	188	8	C	0107-	0207-	0307-	0407-	2007-	3007-	4007-
6	390	10	15	150	188	10	C	0108-	0208-	0308-	0408-	2008-	3008-	4008-
6	470	10	15	150	188	10	C	0109-	0209-	0309-	0409-	2009-	3009-	4009-
6	470	20	15	150	188	10	C	0110-	0210-	0310-	0410-	2010-	3010-	4010-
6	680	10	20	200	250	10	D	0111-	0211-	0311-	0411-	2011-	3011-	4011-
6	680	20	20	200	250	10	D	0112-	0212-	0312-	0412-	2012-	3012-	4012-
6	820	10	20	200	250	10	D	0113-	0213-	0313-	0413-	2013-	3013-	4013-
6	1,000	10	30	300	375	10	D	0114-	0214-	0314-	0414-	2014-	3014-	4014-
6	1,000	20	30	300	375	10	D	0115-	0215-	0315-	0415-	2015-	3015-	4015-
10	6.8	10	1	10	12.5	6	A	0116-	0216-	0316-	0416-	2016-	3016-	4016-
10	6.8	20	1	10	12.5	6	A	0117-	0217-	0317-	0417-	2017-	3017-	4017-
10	8.2	10	1.2	12	15	6	A	0118-	0218-	0318-	0418-	2018-	3018-	4018-
10	47	10	5	50	63	6	B	0119-	0219-	0319-	0419-	2019-	3019-	4019-
10	47	20	5	50	63	6	B	0120-	0220-	0320-	0420-	2020-	3020-	4020-
10	56	10	6	60	75	6	B	0121-	0221-	0321-	0421-	2021-	3021-	4021-
10	68	10	7	70	88	6	B	0122-	0222-	0322-	0422-	2022-	3022-	4022-
10	68	20	7	70	88	6	B	0123-	0223-	0323-	0423-	2023-	3023-	4023-
10	82	10	8	80	100	6	B	0124-	0224-	0324-	0424-	2024-	3024-	4024-
10	220	10	15	150	188	8	C	0125-	0225-	0325-	0425-	2025-	3025-	4025-
10	220	20	15	150	188	8	C	0126-	0226-	0326-	0426-	2026-	3026-	4026-
10	270	10	15	150	188	8	C	0127-	0227-	0327-	0427-	2027-	3027-	4027-
10	390	10	20	200	250	10	D	0128-	0228-	0328-	0428-	2028-	3028-	4028-
10	470	10	20	200	250	10	D	0129-	0229-	0329-	0429-	2029-	3029-	4029-
10	470	20	20	200	250	10	D	0130-	0230-	0330-	0430-	2030-	3030-	4030-
10	560	10	30	300	375	10	D	0131-	0231-	0331-	0431-	2031-	3031-	4031-

See footnote at end of table.

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TABLE I. Style CSR23 characteristics and PIN's - Continued.

DC rated voltage	Capacitance (nom)	Capacitance tolerance	DC leakage at			Dissipation factor at -55°C through +125°C	Case size	PIN M39003/03- 1/ failure rate level for (%/1,000 hours)						
			+25°C	+85°C	+125°C			M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	μF	percent	μA	μA	μA	percent								
15	4.7	10	1	10	12.5	4	A	0132-	0232-	0332-	0432-	2032-	3032-	4032-
15	4.7	20	1	10	12.5	4	A	0133-	0233-	0333-	0433-	2033-	3033-	4033-
15	5.6	10	1.3	13	16.5	4	A	0134-	0234-	0334-	0434-	2034-	3034-	4034-
15	33	10	6	60	75	6	B	0135-	0235-	0335-	0435-	2035-	3035-	4035-
15	33	20	6	60	75	6	B	0136-	0236-	0336-	0436-	2036-	3036-	4036-
15	39	10	6	60	75	6	B	0137-	0237-	0337-	0437-	2037-	3037-	4037-
15	150	10	15	150	188	8	C	0138-	0238-	0338-	0438-	2038-	3038-	4038-
15	150	20	15	150	188	8	C	0139-	0239-	0339-	0439-	2039-	3039-	4039-
15	180	10	15	150	188	8	C	0140-	0240-	0340-	0440-	2040-	3040-	4040-
15	220	10	20	200	250	8	D	0141-	0241-	0341-	0441-	2041-	3041-	4041-
15	220	20	20	200	250	8	D	0142-	0242-	0342-	0442-	2042-	3042-	4042-
15	270	10	20	200	250	8	D	0143-	0243-	0343-	0443-	2043-	3043-	4043-
15	330	10	20	200	250	8	D	0144-	0244-	0344-	0444-	2044-	3044-	4044-
15	330	20	20	200	250	8	D	0145-	0245-	0345-	0445-	2045-	3045-	4045-
20	2.7	10	0.8	8	10	4	A	0146-	0246-	0346-	0446-	2046-	3046-	4046-
20	3.3	10	1	10	12.5	4	A	0147-	0247-	0347-	0447-	2047-	3047-	4047-
20	3.3	20	1	10	12.5	4	A	0148-	0248-	0348-	0448-	2048-	3048-	4048-
20	3.9	10	1.2	12	15	4	A	0149-	0249-	0349-	0449-	2049-	3049-	4049-
20	18	10	4	40	50	6	B	0150-	0250-	0350-	0450-	2050-	3050-	4050-
20	22	10	4	40	50	6	B	0151-	0251-	0351-	0451-	2051-	3051-	4051-
20	22	20	4	40	50	6	B	0152-	0252-	0352-	0452-	2052-	3052-	4052-
20	27	10	5	50	63	6	B	0153-	0253-	0353-	0453-	2053-	3053-	4053-
20	56	10	9	90	110	6	C	0154-	0254-	0354-	0454-	2054-	3054-	4054-
20	68	10	10	100	125	6	C	0155-	0255-	0355-	0455-	2055-	3055-	4055-
20	68	20	10	100	125	6	C	0156-	0256-	0356-	0456-	2056-	3056-	4056-
20	82	10	10	100	125	6	C	0157-	0257-	0357-	0457-	2057-	3057-	4057-
20	100	10	15	150	188	6	C	0158-	0258-	0358-	0458-	2058-	3058-	4058-
20	100	20	15	150	188	6	C	0159-	0259-	0359-	0459-	2059-	3059-	4059-
20	120	10	15	150	188	6	C	0160-	0260-	0360-	0460-	2060-	3060-	4060-
20	150	10	20	200	250	8	D	0161-	0261-	0361-	0461-	2061-	3061-	4061-

See footnote at end of table.

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TABLE I. Style CSR23 characteristics and PIN's - Continued.

DC rated voltage	Capacitance (nom)	Capacitance tolerance	DC leakage at			Dissipation factor at -55°C through +125°C	Case size	PIN M39003/03- 1/ failure rate level for (%/1,000 hours)						
			+25°C	+85°C	+125°C			M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	μF	percent	μA	μA	μA	percent								
20	150	20	20	200	250	8	D	0162-	0262-	0362-	0462-	2062-	3062-	4062-
20	180	10	20	200	250	8	D	0163-	0263-	0263-	0463-	2063-	3063-	4063-
35	1.8	10	1	10	12.5	4	A	0164-	0264-	0364-	0464-	2064-	3064-	4064-
35	8.2	10	3.5	35	44	6	B	0165-	0265-	0365-	0465-	2065-	3065-	4065-
35	10	10	4	40	50	6	B	0166-	0266-	0366-	0466-	2066-	3066-	4066-
35	10	20	4	40	50	6	B	0167-	0267-	0367-	0467-	2067-	3067-	4067-
35	33	10	10	100	125	6	C	0168-	0268-	0368-	0468-	2068-	3068-	4068-
35	33	20	10	100	125	6	C	0169-	0269-	0369-	0469-	2069-	3069-	4069-
35	39	10	10	100	125	6	C	0170-	0270-	0370-	0470-	2070-	3070-	4070-
35	47	10	10	100	125	6	C	0171-	0271-	0371-	0471-	2071-	3071-	4071-
35	47	20	10	100	125	6	C	0172-	0272-	0372-	0472-	2072-	3072-	4072-
35	56	10	15	150	188	6	D	0173-	0273-	0373-	0473-	2073-	3073-	4073-
35	68	10	15	150	188	6	D	0174-	0274-	0374-	0474-	2074-	3074-	4074-
35	68	20	15	150	188	6	D	0175-	0275-	0375-	0475-	2075-	3075-	4075-
50	1.2	10	0.9	9	11	4	A	0176-	0276-	0376-	0476-	2076-	3076-	4076-
50	1.5	10	1.2	12	15	4	A	0177-	0277-	0377-	0477-	2077-	3077-	4077-
50	1.5	20	1.2	12	15	4	A	0178-	0278-	0378-	0478-	2078-	3078-	4078-
50	5.6	10	4.5	45	56	4	B	0179-	0279-	0379-	0479-	2079-	3079-	4079-
50	6.8	10	4.5	45	56	6	B	0180-	0280-	0380-	0480-	2080-	3080-	4080-
50	6.8	20	4.5	45	56	6	B	0181-	0281-	0381-	0481-	2081-	3081-	4081-
50	22	10	10	100	125	6	C	0182-	0282-	0382-	0482-	2082-	3082-	4082-
50	22	20	10	100	125	6	C	0183-	0283-	0383-	0483-	2083-	3083-	4083-
50	27	10	10	100	125	6	C	0184-	0284-	0384-	0484-	2084-	3084-	4084-
50	33	10	10	100	125	6	D	0185-	0285-	0385-	0485-	2085-	3085-	4085-
50	33	20	10	100	125	6	D	0186-	0286-	0386-	0486-	2086-	3086-	4086-
50	39	10	10	100	125	6	D	0187-	0287-	0387-	0487-	2087-	3087-	4087-

1/ Dash number shall include the letter "H" to indicate an internal construction solder with a minimum melting point of 221°C, or the "H" shall be deleted.

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REQUIREMENTS - Continued.

Stability at low and high temperatures: In accordance with MIL-PRF-39003.

Step 1 (+25◦C):

DCL: See table I.

Cap.: Within tolerance of table I value.

DF: See table I.

Step 2 (-55◦C):

ΔCap.: Shall not change more than ±10 percent of step 1 measured value.

DF: See table I.

Step 3 (+25◦C):

DCL: See table I.

ΔCap.: Shall not change more than ±2 percent of step 1 measured value.

DF: See table I.

Step 4 (+85◦C):

DCL: See table I.

ΔCap.: Shall not change more than ±8 percent of step 1 measured value.

DF: See table I.

Step 5 (+125◦C):

DCL: See table I.

ΔCap.: Shall not change more than ±12 percent of step 1 measured value.

DF: See table I.

Step 6 (+25◦C):

DCL: See table I.

ΔCap.: Shall not change more than ±2 percent of step 1 measured value.

DF: See table I.

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Surge voltage: In accordance with MIL-PRF-39003.

DCL: See table I.

ΔCap.: Shall not change more than ±2 percent of initial measured value.

DF: See table I.

Life: Method 108 of MIL-STD-202.

240, 1,000, and 2,000 hours:

At +25°C (as applicable):

DCL: See table I.

ΔCap.: Shall not change more than ±5 percent of initial measured value.

DF: See table I.

At +85°C:

DCL: See table I.

At +125°C:

DCL: See table I.

4,000, 6,000, 8,000, and 10,000 hours (exponential only):

At +25°C (as applicable):

DCL: 200 percent of initial limit.

ΔCap.: Shall not change more than ±10 percent of initial measured value.

DF: See table I.

At +85°C:

DCL: 200 percent of initial limit.

At +125°C:

DCL: 200 percent of initial limit.

Marking: In accordance with MIL-PRF-39003.

Part or Identifying Number (PIN): M39003/03- (dash number from table I).

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Substitutability data: Substitutability for failure rate levels and capacitance tolerance shall be in accordance with MIL-PRF-39003.

* CHANGES FROM PREVIOUS ISSUE: The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* Referenced documents. In addition to MIL-PRF-39003, this document also references MIL-STD-202.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2257)

Review activities:
Navy - AS, MC, OS, SH
Air Force - 19, 99

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil

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